

Anastasiia Mokhonko

DATA SCIENCE & AI STUDENT

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"Aspiring Data Scientist and AI Specialist passionate about digital innovation, machine learning, and solving real-world problems."

Summary

I am a second-year student at Breda University of Applied Sciences, where I have developed a solid foundation in data science, artificial intelligence, and applied machine learning. Throughout my studies, I have gained hands-on experience with Python, SQL, Power BI, and industry-standard tools for data analysis and visualization. I have built and deployed end-to-end machine learning pipelines, worked with deep learning frameworks (including transformers and U-Net), and applied NLP and computer vision to real-world projects. My academic journey also emphasized team-work, agile project management, scientific research, and the ethical and legal aspects of AI, such as GDPR and the EU AI Act. I am passionate about solving complex problems and continuously expanding my technical and analytical skills.

Experience

Emotion Detection NLP Pipeline – Content Intelligence Agency (Phase 2)

Breda, NL

MLOPS ENGINEER, DEVELOPER

May 2025 – Jun 2025

- Deployed production-ready emotion detection system classifying 6 emotions in French video/audio content with 44.7 requests/second capacity.
- Built Azure ML pipelines with automated training, retraining, and deployment using Docker containers and CI/CD integration.
- Developed secure FastAPI endpoints and Streamlit interface for multi-user access with monitoring and logging.
- Implemented blue-green deployment strategy with model versioning and rollback capabilities for zero-downtime updates.

Emotion Detection from Video – Content Intelligence Agency (Phase 1)

Breda, NL

DEVELOPER, NLP ENGINEER

Feb 2025 – Apr 2025

- Developed NLP pipeline for emotion analysis in French reality TV including audio transcription, translation, and classification.
- Implemented and compared Naive Bayes, Logistic Regression, LSTM, BERT, and CamemBERT models for emotion detection.
- Applied XAI techniques (LIME, SHAP) for model interpretability; delivered research documentation and model card to client.

Root Segmentation & Robotic Control – NPEC Netherlands Plant Eco-phenotyping Centre

Breda, NL

DEVELOPER, COMPUTER VISION ENGINEER

Nov 2024 – Dec 2024

- Built computer vision pipeline using U-Net for plant root segmentation and graph theory (NetworkX, Dijkstra) for measurement.
- Developed PID and Reinforcement Learning controllers (Stable Baselines 3) for Opentrons OT-2 robotic microbial application.
- Achieved 99% time reduction in root analysis workflow; delivered high-accuracy solution to research client.

Cybersecurity Awareness in Small Hotels – Policy Research for DigiWerkplaats

Breda, NL

ANALYTICS TRANSLATOR, TEAM LEAD, RESEARCHER

Sep 2024 – Oct 2024

- Led research team of 5 investigating employee cybersecurity awareness impact on hotel security using mixed-methods approach.
- Designed and analyzed surveys (Qualtrics) and interviews; ensured GDPR compliance and research ethics approval.
- Found moderate positive correlation ($r=0.456$) between training frequency and security adherence; recommended quarterly training.

ANWB Safe Driving Application – ANWB Dutch Automobile Association

Breda, NL

DEVELOPER, DATA ANALYST

May 2024 – Jun 2024

- Developed Streamlit application analyzing driver behavior and road safety using ANWB datasets and time series analysis.
- Implemented incident classification models with EU AI Act compliance and Agile project management methodology.

Farmer's Buddy: Potato Blight Detection AI

Breda, NL

DEVELOPER, DATA SCIENTIST

Feb 2024 – Apr 2024

- Built AI-driven mobile application for potato blight detection using MobileNetV2 transfer learning, achieving 95.3% accuracy.
- Applied Grad-CAM for explainable AI and conducted think-aloud usability studies to optimize user experience.

Goal Prediction Analytics – NAC Breda Football Club

Breda, NL

DATA ANALYST, MACHINE LEARNING ENGINEER

Nov 2023 – Jan 2024

- Developed machine learning model analyzing 16,535 players to predict goal-scoring potential with 95%+ accuracy.
- Delivered GDPR-compliant recruitment recommendations using linear regression and correlation analysis.

Human Trafficking Data Analysis Dashboard – UN SDG 16

Breda, NL

DEVELOPER, DATA ANALYST

Sep 2023 – Nov 2023

- Designed interactive Power BI dashboard analyzing global trafficking patterns by age, gender, exploitation type, and region.
- Calculated temporal correlation ($r=0.68$) and delivered policy recommendations for prevention strategies supporting SDG 16.

Education

Breda University of Applied Sciences
BACHELOR OF APPLIED DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Breda, Netherlands
Sep 2023 – Present

ITSTEP Academy
PYTHON DEVELOPER COURSE

Kyiv, Ukraine
2022 – 2023

- Completed 6-month intensive Python programming and backend development course.
- Covered Python fundamentals, OOP, data structures, algorithms, SQL/NoSQL databases, and Django web development.
- Gained hands-on experience in asynchronous programming, software testing, Git version control, and cloud technologies.

Extracurricular Activity

Degree Programme Committee (DPC) – Data Science & AI Programme

Breda University of Applied Sciences, Breda, NL
2024 – Present

- Selected as one of three student representatives to serve on the official programme committee alongside faculty members.
- Collaborate with lecturers and programme management to improve curriculum quality, course delivery, and student experience.
- Propose ideas for programme enhancement, address student concerns, and help resolve issues encountered by both students and staff.
- Participate in regular committee meetings to review programme policies, assessment methods, and educational standards.
- Act as liaison between student body and faculty, ensuring student perspectives are represented in programme decisions.

Nakormim Bali – Orphanage Support Organization

Bali, Indonesia
2023 – Present

- Support post-COVID-19 pandemic relief efforts for orphanages, helping to feed children and facilitate adoption processes to find them forever homes.
- Provide ongoing monthly financial donations to sustain orphanage operations and child welfare programs.
- Organized educational study sessions for children, creating supportive learning environments and mentoring opportunities.
- Coordinated group excursions including beach trips, turtle hatching observation events, and creative activities such as pottery workshops.
- Contribute to holistic child development through recreational, educational, and cultural experiences.

Skills

Python, HTML, SQL, Git, Power BI, Figma, Streamlit, VSCode 5

Machine Learning, Deep Learning, NLP, Computer Vision, Data Visualization, XAI 5

Languages

Russian (Native) 5

Ukrainian (Native) 5

English (C1) 4

French (B2) 4